ASSIGNMENT-1

- **PROBLEM STATEMENT**

  Given the 5-variable Boolean function:F (A, B, C, D, E) = Σm(2, 3, 6, 7, 8, 9, 12, 13).Use AI assistance to obtain the minimized expression and verify its functionality.

- **OBJECTIVE**

AI-Assisted Boolean Expression Simplifier — use GenAI to reduce a 4–6 variable expression (SOP/POS, K-map/algebraic), then implement and verify the simplified circuit in a simulator.

- **USING GEN AI FOR SIMPLIFICATION**

  AI PROMPT

  Simplify a 4-variable Boolean expression in Sum of Products (SOP) form: F (A, B, C, D) = Σm(2, 3, 6, 7, 8, 9, 12, 13). Please use a Karnaugh map (K-map) and algebraic minimization to reduce this expression to its simplest form. Show your step-by-step grouping and provide the final minimized Boolean equation.

  AI RESULT

  To simplify the Boolean expression $F(A, B, C, D) = \sum m(2, 3, 6, 7, 8, 9, 12, 13)$, we map the given minterms onto a 4-variable Karnaugh map.Minterms translate to their binary equivalents:

- **m2**: 0010, **m3**: 0011, **m6**: 0110, **m7**: 0111
- **m8**: 1000, **m9**: 1001, **m12**: 1100, **m13**: 1101

1. Karnaugh Map Plotting

Place a **1** in the cells corresponding to these minterms and a **0** (or leave blank) in the rest.

AB \ CD	**00 ($C'D'$)**	**01 ($C'D$)**	**11 (CD)**	**10 (CD')**
00 ($A'B'$)	0	0	**1** (m3)	**1** (m2)
01 ($A'B$)	0	0	**1** (m7)	**1** (m6)
11 (AB)	**1** (m12)	**1** (m13)	0	0
10 (AB')	**1** (m8)	**1** (m9)	0	0

2. Step-by-Step Grouping

We group the 1s into the largest possible blocks (powers of 2). Here, we can form two distinct groups of four (quads).

Group 1 (Top-Right Quad):

- **Minterms covered:** m2, m3, m6, m7
- **Rows spanned:** 00 and 01. Variable A remains 0 (A'), while B changes from 0 to 1 and cancels out.
- **Columns spanned:** 11 and 10. Variable C remains 1 (C), while D changes from 1 to 0 and cancels out.
- **Extracted Term:** $A'C$

Group 2 (Bottom-Left Quad):

- **Minterms covered:** m8, m9, m12, m13
- **Rows spanned:** 11 and 10. Variable A remains 1 (A), while B changes from 1 to 0 and cancels out.
- **Columns spanned:** 00 and 01. Variable C remains 0 (C'), while D changes from 0 to 1 and cancels out.
- **Extracted Term:** AC'

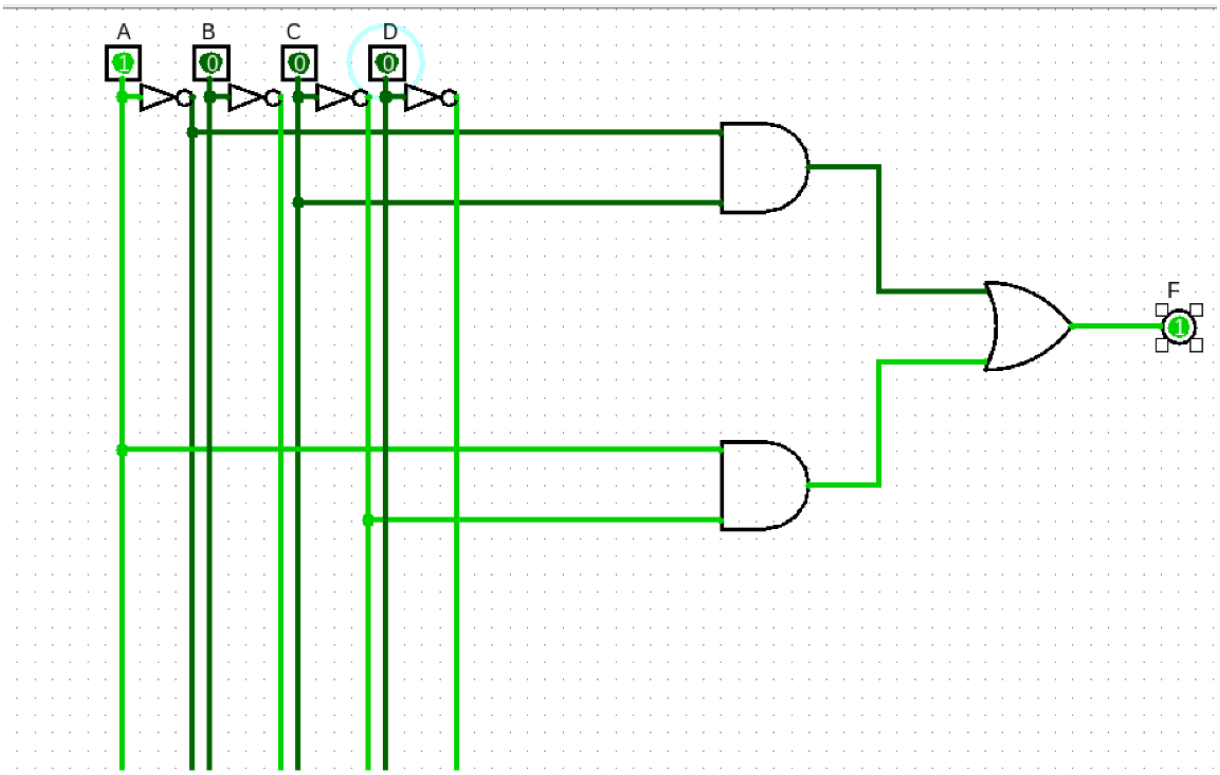
3. Final Minimized Expression

Combine the extracted terms using the OR operator (Sum of Products).

$F = A \cdot C + A C \cdot$

(Note: This expression represents an Exclusive-OR (XOR) gate for variables A and C, which can also be written as $F = A \oplus C$.)

- **Circuit Implementation & Logisim Simulation**



- ## Discussion and Verification

- **Truth Table Verification:** To verify the simplified circuit in the simulator, cycle through all 16 possible input combinations (0000 to 1111) for A, B, C, and D in Logisim. The output pin F will light up (output 1) exactly on the decimal combinations 2, 3, 6, 7, 8, 9, 12, and 13. For example, if A=1, B=0, C=0, D=1 (minterm 9), A=1 and C=0 makes the AC' term True, resulting in F=1. This perfectly matches the original unsimplified specification.

- **Reflection on GenAI's Output:** The GenAI effectively demonstrated how a 4-variable expression can be drastically reduced. By using K-map grouping, the AI showed that variables B and D are redundant in these specific minterm clusters. Building the unsimplified circuit would have required an 8-input OR gate and eight 4-input AND gates. The AI-assisted reduction minimized this to just two AND gates, one OR gate, and two inverters, significantly optimizing hardware cost and propagation delay.